



May 2025

PH1003-1 – PHYSICS OF FLUIDS AND SEMICONDUCTORS
(For BT)

Max. Marks: 100

Physical Constants: Velocity of light, $c=3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h=6.63 \times 10^{-34} \text{ Js}$,
Electron mass, $m=9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e=1.6 \times 10^{-19} \text{ C}$,
Boltzmann constant, $k=1.38 \times 10^{-23} \text{ J/K}$. Avogadro number, $N_A = 6.022 \times 10^{26} / \text{kg mole}$.

20 Marks

- Which of the following is a characteristic of a fluid?
A) Fixed shape
B) Ability to flow
C) Incompressible
D) Rigid structure
- Pressure is defined as force per unit _____.
A) Volume
B) Area
C) Mass
D) Density
- In a static fluid, pressure increases with:
A) Increase in temperature
B) Decrease in depth
C) Increase in depth
D) Increase in viscosity
- The density of an object submerged in water determines its:
A) Buoyancy
B) Pressure
C) Volume
D) Weight
- Archimedes' principle applies to:
A) Liquids only
B) Gases only
C) Both liquids and gases
D) Solids only
- The continuity equation is derived from the principle of:
A) Conservation of energy
B) Conservation of mass
C) Conservation of momentum
D) Newton's Second Law
- Bernoulli's equation is based on the principle of:
A) Conservation of mass
B) Conservation of energy
C) Conservation of momentum
D) Newton's third law
- If the Reynolds number is very high, the fluid flow is
A) Laminar
B) Turbulent
C) Steady
D) Stationary
- Which particles are responsible for electrical conduction in metals?
A) Protons
B) Neutrons
C) Electrons
D) Ions
- What is the mean free path of an electron?
A) The shortest distance an electron can travel
B) The average distance an electron travels between collisions
C) The maximum distance an electron moves in a metal
D) The distance an electron travels in a vacuum

11. What is the Fermi energy in metals?
 - A) The total energy of an atom
 - B) The highest energy level occupied by electrons at absolute zero temperature
 - C) The minimum energy needed for an electron to escape a metal
 - D) The energy at which electrons are at rest
12. What happens to the electrical resistivity of metals as temperature increases?
 - A) It decreases
 - B) It remains constant
 - C) It increases
 - D) It first increases, then decreases
13. Which rule explains the effect of impurities on the resistivity of metals?
 - A) Nordheim's rule
 - B) Matthiessen's rule
 - C) Ohm's law
 - D) Kirchhoff's law
14. Which type of semiconductor is formed by adding pentavalent impurities?
 - A) n-type semiconductor
 - B) p-type semiconductor
 - C) Intrinsic semiconductor
 - D) Insulator
15. The Hall effect is used to determine:
 - A) The type of charge carriers in a material
 - B) The speed of sound in a solid
 - C) The energy levels in an atom
 - D) The density of protons in a metal
16. Which device converts light energy into electrical energy?
 - A) LED
 - B) Photodiode
 - C) Solar cell
 - D) Diode
17. Which law is fundamental for determining the diffraction angle in X-ray diffraction?
 - A) Newton's Law
 - B) Bragg's Law
 - C) Planck's Law
 - D) Faraday's Law
18. Which of the following lenses are used in an electron microscope?
 - A) Electrostatic and magnetostatic lenses
 - B) Optical lenses
 - C) Convex and concave lenses
 - D) Objective lenses
19. Which type of electron microscope is used to view the surface of a sample in high resolution?
 - A) Scanning Electron Microscope (SEM)
 - B) Transmission Electron Microscope (TEM)
 - C) Atomic Force Microscope (AFM)
 - D) Optical microscope
20. Which of the following is a common application of mass spectrometry?
 - A) Identifying chemical compounds based on mass-to-charge ratio
 - B) Measuring temperature
 - C) Measuring the wavelength of X-rays
 - D) Generating atomic spectra

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1.	a) Explain the concept of fluid statics and fluid dynamics with suitable examples.	4	L*1	1	1,2
	b) Explain the principle of buoyancy and derive the expression for the buoyant force using Archimedes' principle.	8	L2	1	1,2
	c) A wooden cube of volume 250 cm^3 weighs 3 kgf in air. When fully immersed in a liquid of density $1.2 \times 10^3 \text{ kg/m}^3$, what is the upthrust experienced?	4	L3	1	1,2
2.	a) Discuss the concepts of absolute pressure, gauge pressure, and atmospheric pressure with examples.	4	L1, L2	1	1,2
	b) What is a Pitot's tube? Explain the principle of operation of a Pitot tube and how it is used to measure fluid velocity. Discuss its advantages and limitations.	8	L2	2	1,2
	c) A hydraulic system has a small piston with a radius of 2 cm. If a force of 80 N is applied to the piston, determine the pressure exerted on the fluid.	4	L3	1	1,2

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|-------|---|---|-----------|---|------|
| 3. a) | Explain the significance of the Reynolds number in fluid flow. | 4 | L1,
L2 | 2 | 1,2 |
| b) | What is the equation of continuity? Derive the mathematical form of the continuity equation for incompressible fluids. | 8 | L1,
L2 | 2 | 1,2 |
| c) | A metal block is suspended from a spring balance. It weighs 40 N in air, 28 N when fully immersed in water, and 32 N when immersed in another liquid of unknown density. Determine the density of the unknown liquid. | 4 | L3 | 2 | 1,2. |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Explain the drawbacks of classical free electron theory. | 4 | L1 | 3 | 1,2 |
| b) | Based on classical free electron theory, derive the expression for the electrical conductivity of metals. | 8 | L2 | 3 | 1,2 |
| c) | Determine the electron mobility in aluminum given that its resistivity is $2.65 \times 10^{-8} \Omega \cdot m$ and its free electron density is 6.02×10^{28} electrons per cubic meter. | 4 | L3 | 3 | 1,2 |
| 5. a) | State and explain Matthiessen's rule. | 4 | L1 | 3 | 1,2 |
| b) | What is the Hall effect? Explain the production of the Hall field and obtain an expression for the Hall coefficient and carrier concentration of an n-type semiconductor. | 8 | L2 | 4 | 1,2 |
| c) | A n-type semiconductor has a Hall coefficient of $3.66 \times 10^{-4} m^3/C$ and its resistivity is 2.12 ohm m. Calculate charge carrier concentration and electron mobility at room temperature. | 4 | L3 | 4 | 1,2 |
| 6. a) | What are intrinsic & extrinsic semiconductors? Mention the position of the Fermi level in intrinsic, n-type, and p-type semiconductors at 0 K. | 6 | L1 | 4 | 1,2 |
| b) | Explain the construction and working of solar cell. | 6 | L2 | 4 | 1,2 |
| c) | Calculate the probability of occupation of an electron occupying an energy level 0.04 eV above the Fermi level at 400 K. | 4 | L3 | 4 | 1,2 |

Unit – III

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|-------|---|---|-----------|---|-----|
| 7. a) | Explain in detail the difference between electrostatic and magnetostatic lenses. | 6 | L1,
L2 | 5 | 1,2 |
| b) | Explain the basic construction and working of an SEM. | 6 | L2 | 5 | 1,2 |
| c) | A particle with a charge of $2.00 \times 10^{-19} C$ and mass of $9.11 \times 10^{-31} kg$ is released from rest in a uniform electric field of magnitude $1.50 \times 10^4 N/C$. Find its acceleration. | 4 | L3 | 5 | 1,2 |
| 8. a) | Explain the origin of characteristic X-rays with appropriate diagrams. | 6 | L2 | 5 | 1,2 |
| b) | With a neat diagram explain the principle and working of a mass spectrometer. | 6 | L2 | 5 | 1,2 |
| c) | A crystal with a cubic structure is analyzed using X-ray diffraction with a wavelength of 1.6 Å. The first-order reflection from the (111) plane is observed at an angle of 55° . Calculate the lattice parameter. | 4 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
